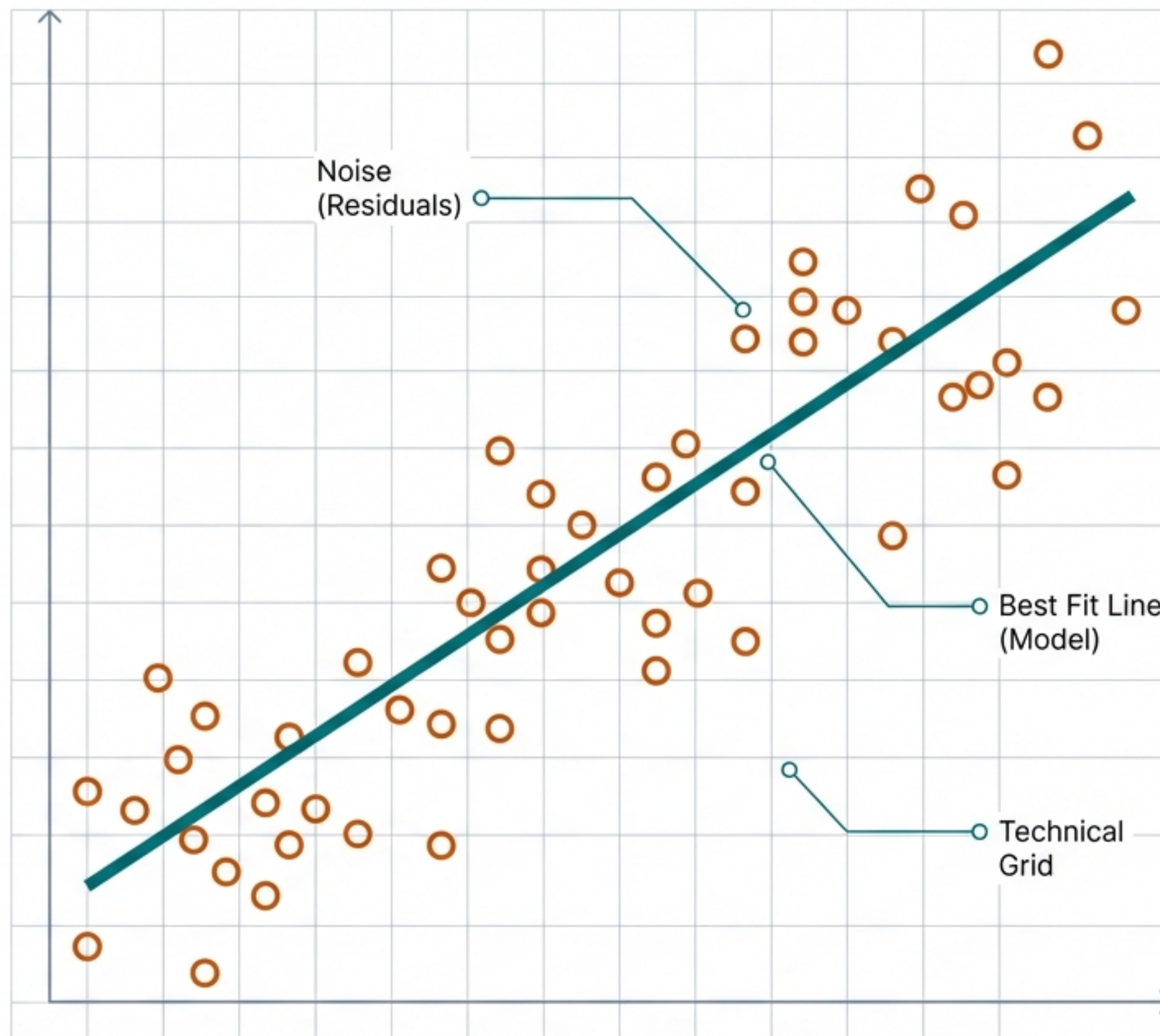


Linear Regression: The Art of the Best Fit

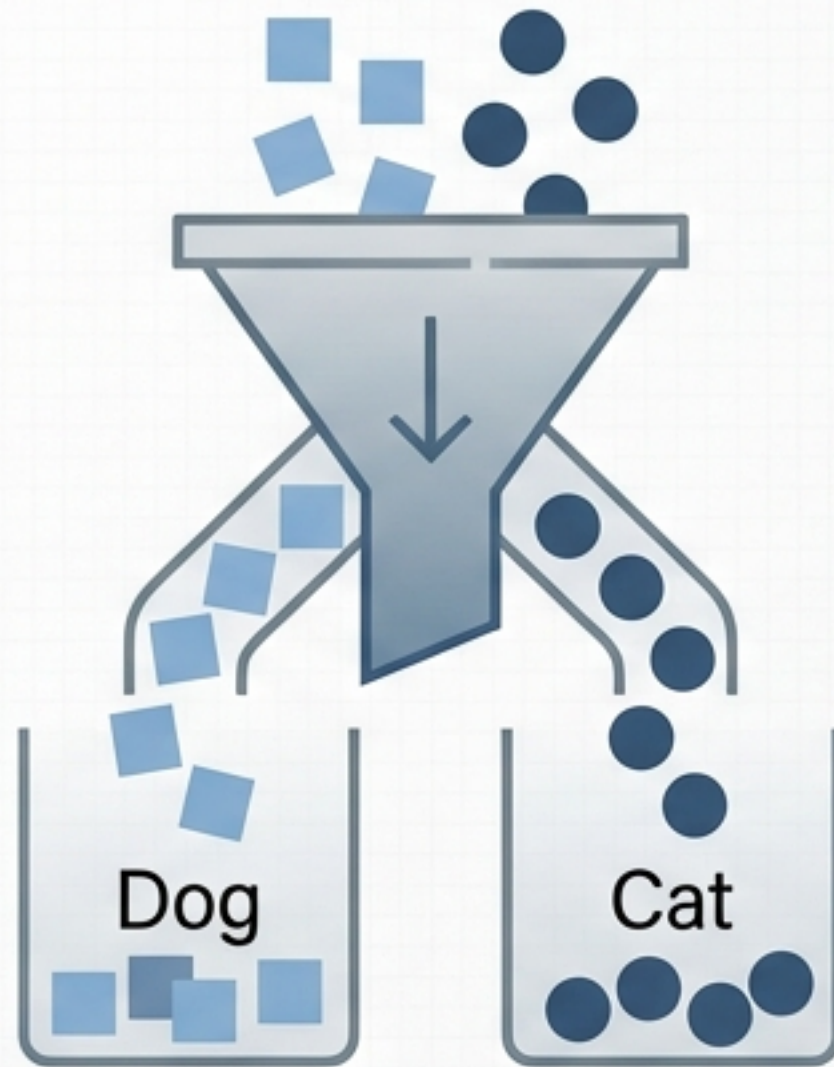
From Geometric Intuition to
Python Implementation.



Based on the methodology of Codanics

THE OUTPUT DEFINES THE METHOD

CLASSIFICATION



Categorical Output

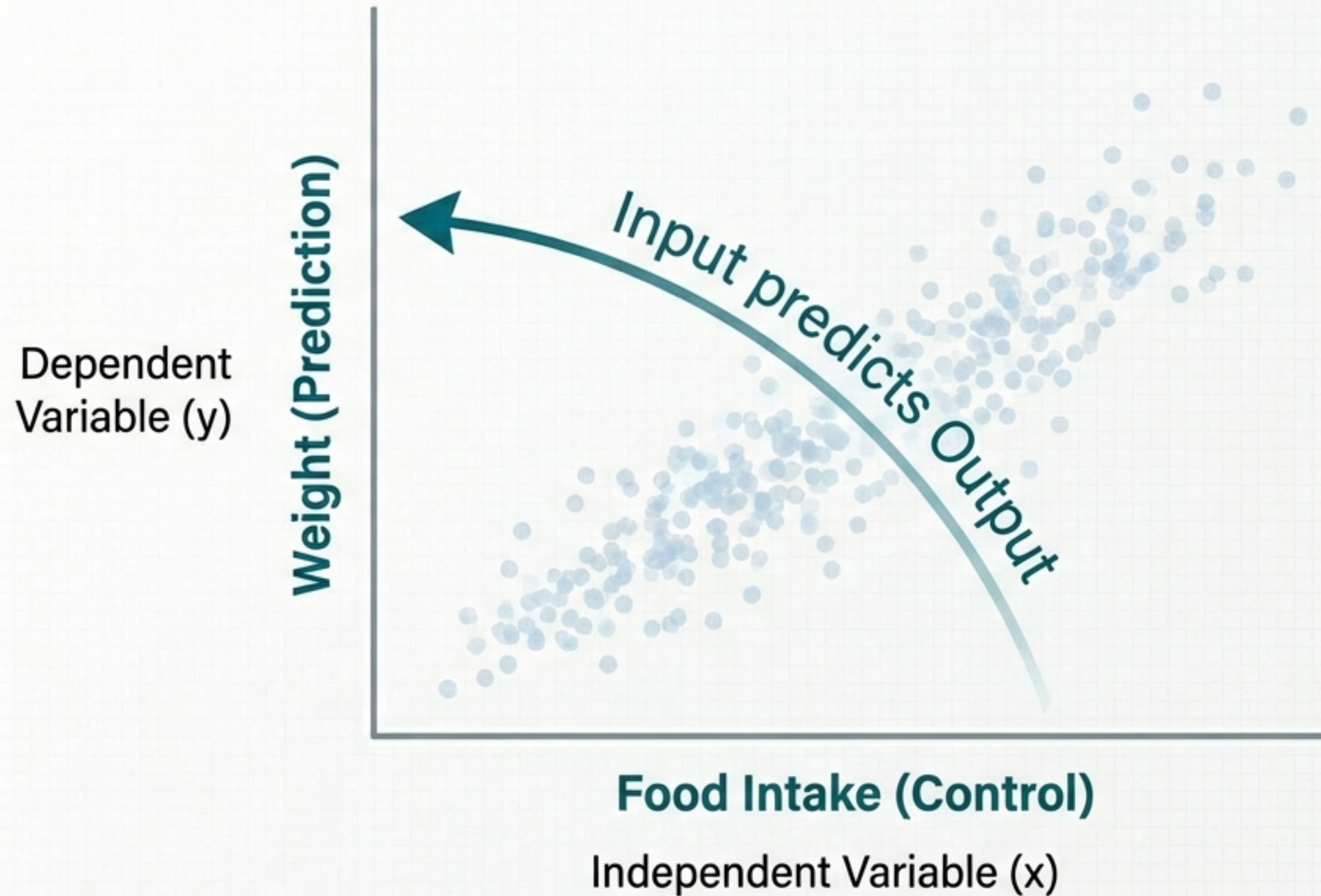
REGRESSION



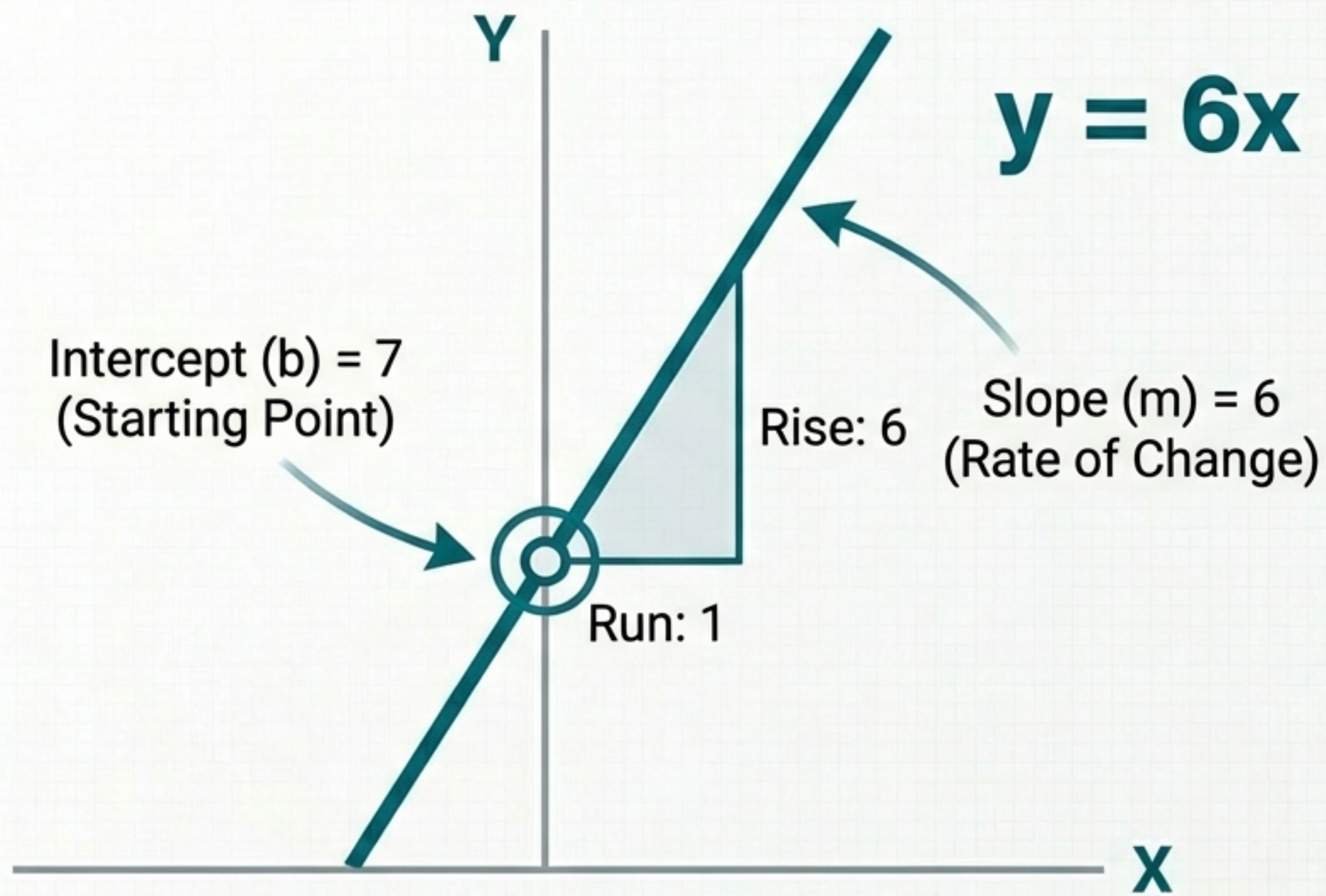
Numeric/Continuous Output

Key Insight: If the target (y) is a continuous number, the problem requires Regression.

THE ANATOMY OF A RELATIONSHIP



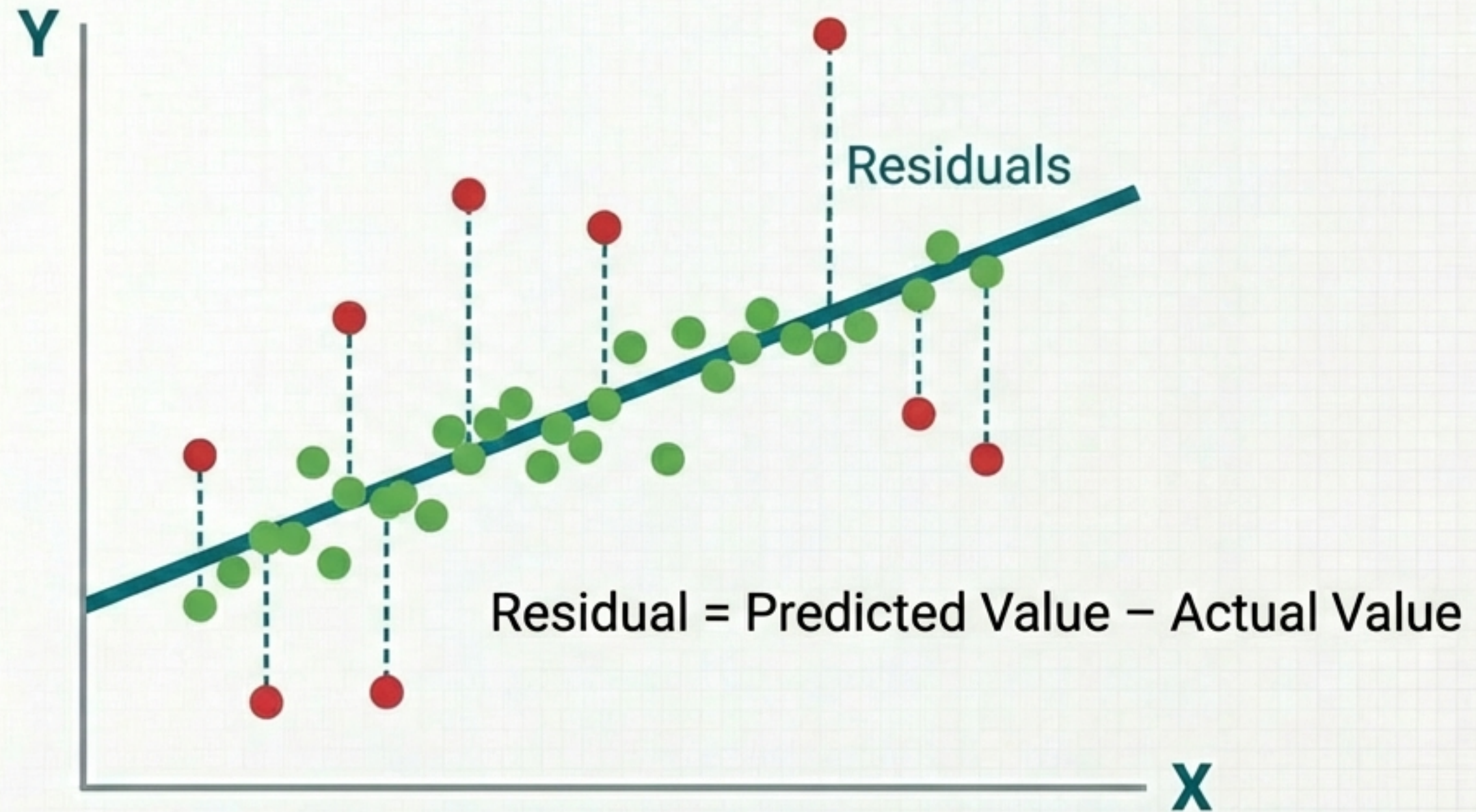
DRAWING THE LINE ($y = mx + b$)



Example Calculation:

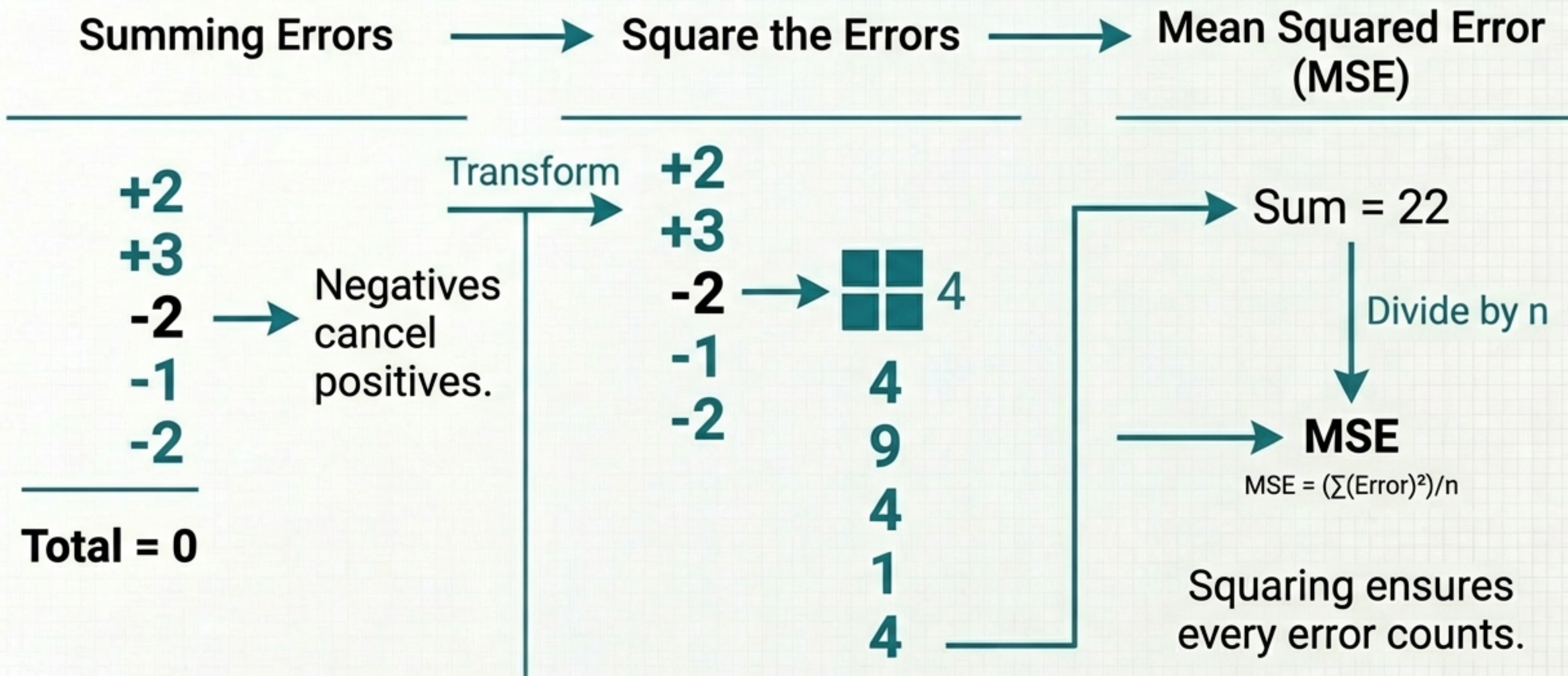
$$\begin{aligned}\text{If Input } (x) &= 2 \\ y &= 6(2) + 7 \\ y &= 19\end{aligned}$$

THE PROBLEM OF “UNHAPPY” POINTS

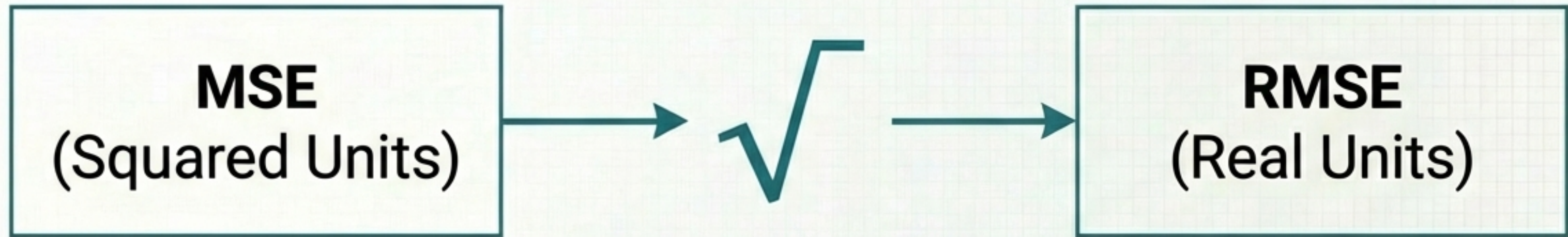


The line is a compromise. The goal is to minimize the vertical distance (unhappiness) for the entire group simultaneously.

Quantifying the Error (MSE)



Returning to Reality (RMSE)



MSE is abstract (e.g., 'squared kilograms').
RMSE is practical (e.g., 'The model is off by $\pm 5\text{kg}$ ').

"It specifies the absolute fit of the model to the data."

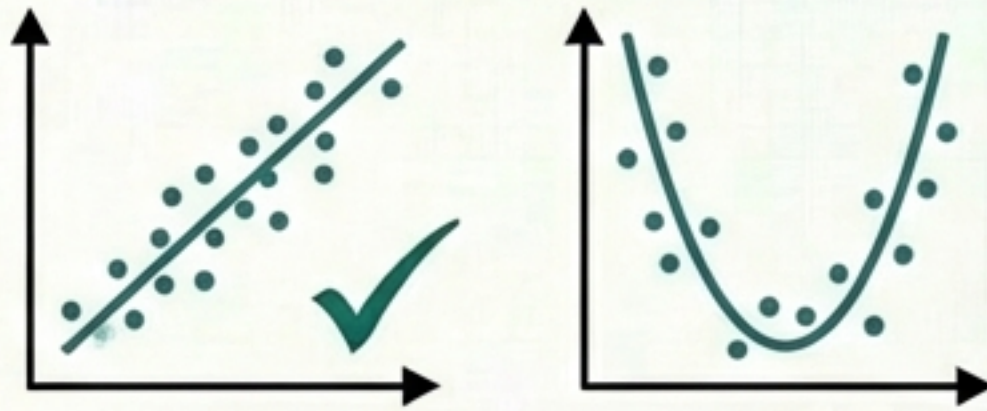
The Model's Scorecard (R^2)



R^2 represents the percentage of variance in the data explained by the model.

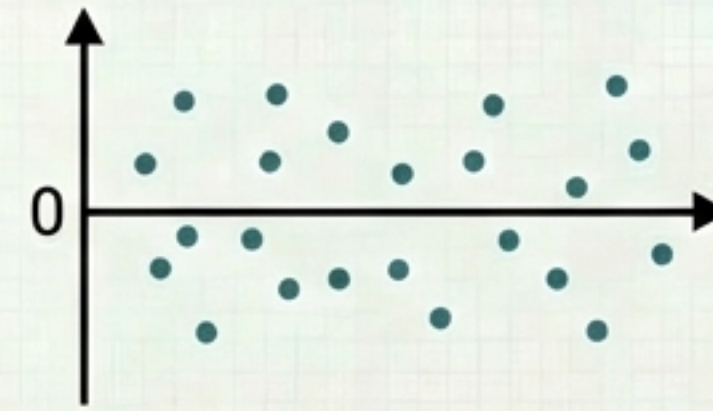
The Assumptions Checklist

Linearity



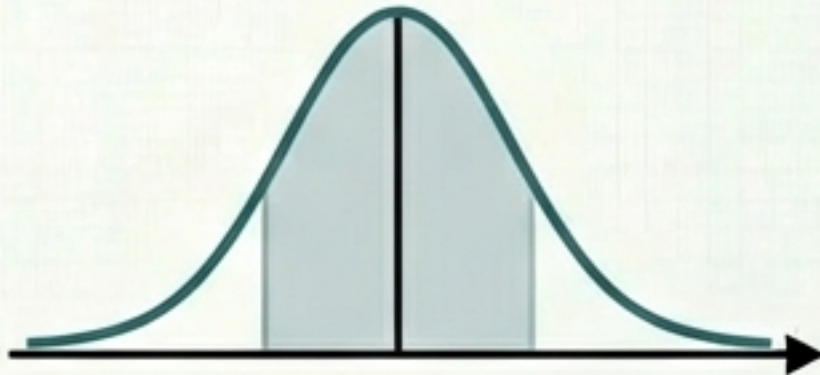
Relationship must be linear.

Independence



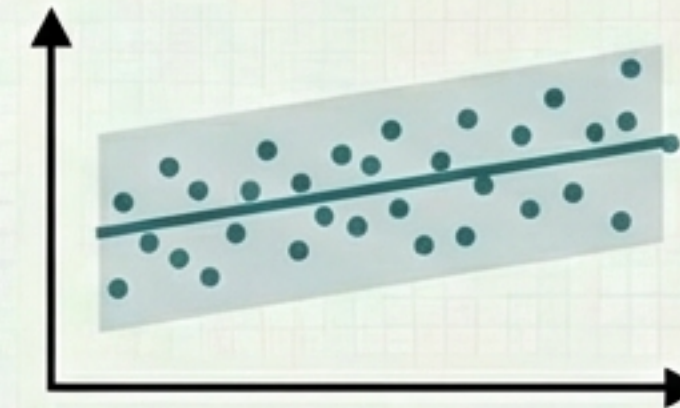
Residuals must be independent.

Normality



Residuals follow a normal distribution.

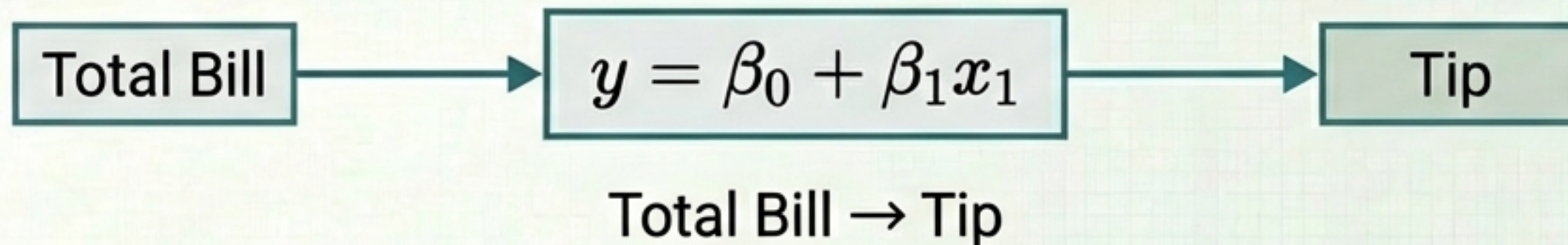
Homoscedasticity



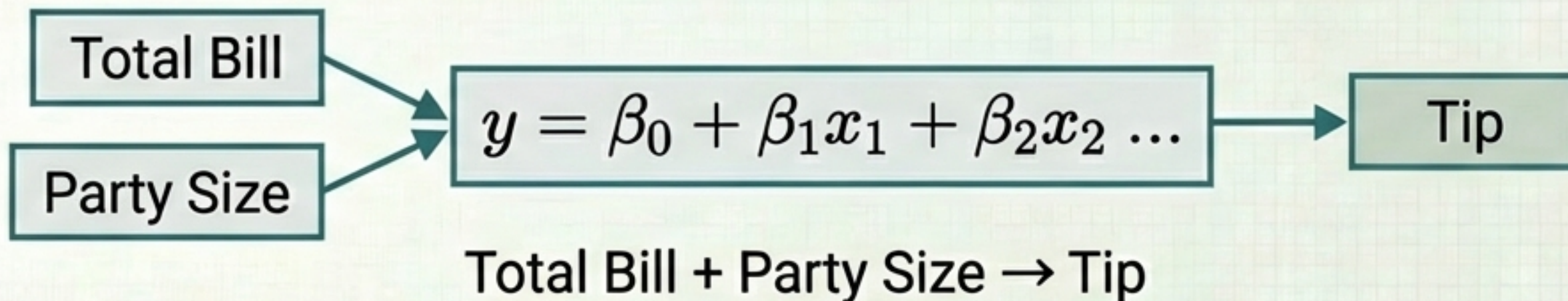
Variance remains consistent.

From Simple to Multi-Linear

Simple Regression (1 Input)

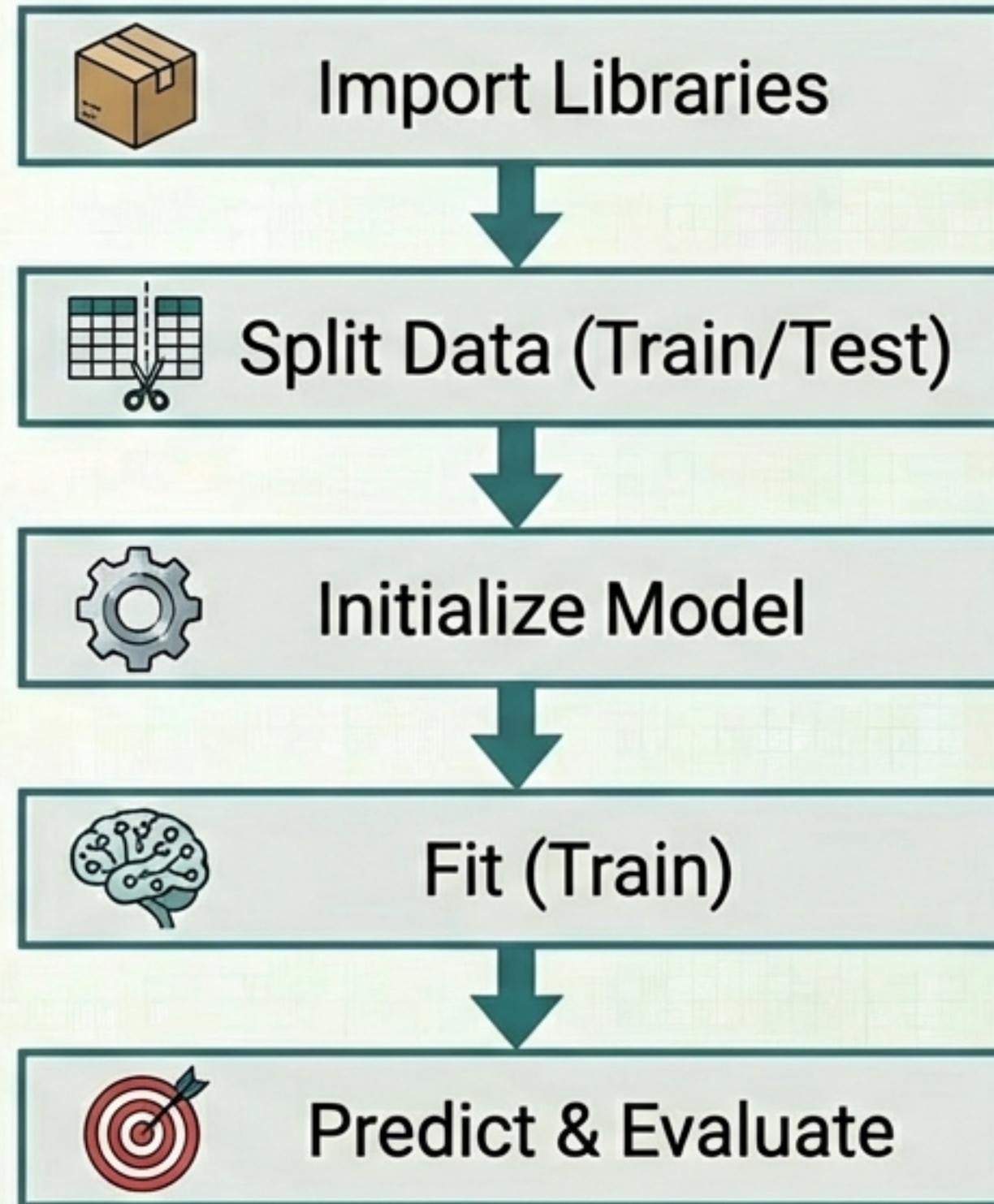


Multi-Linear Regression (Multiple Inputs)



Adding more relevant inputs helps the model capture more complexity.

The Scikit-Learn Blueprint



Step 1 & 2: Preparation & Split

```
import pandas as pd
from sklearn.model_selection import
    train_test_split

# 80/20 Rule
X_train, X_test, y_train, y_test =
    train_test_split(X, y,
        test_size=0.2)
```

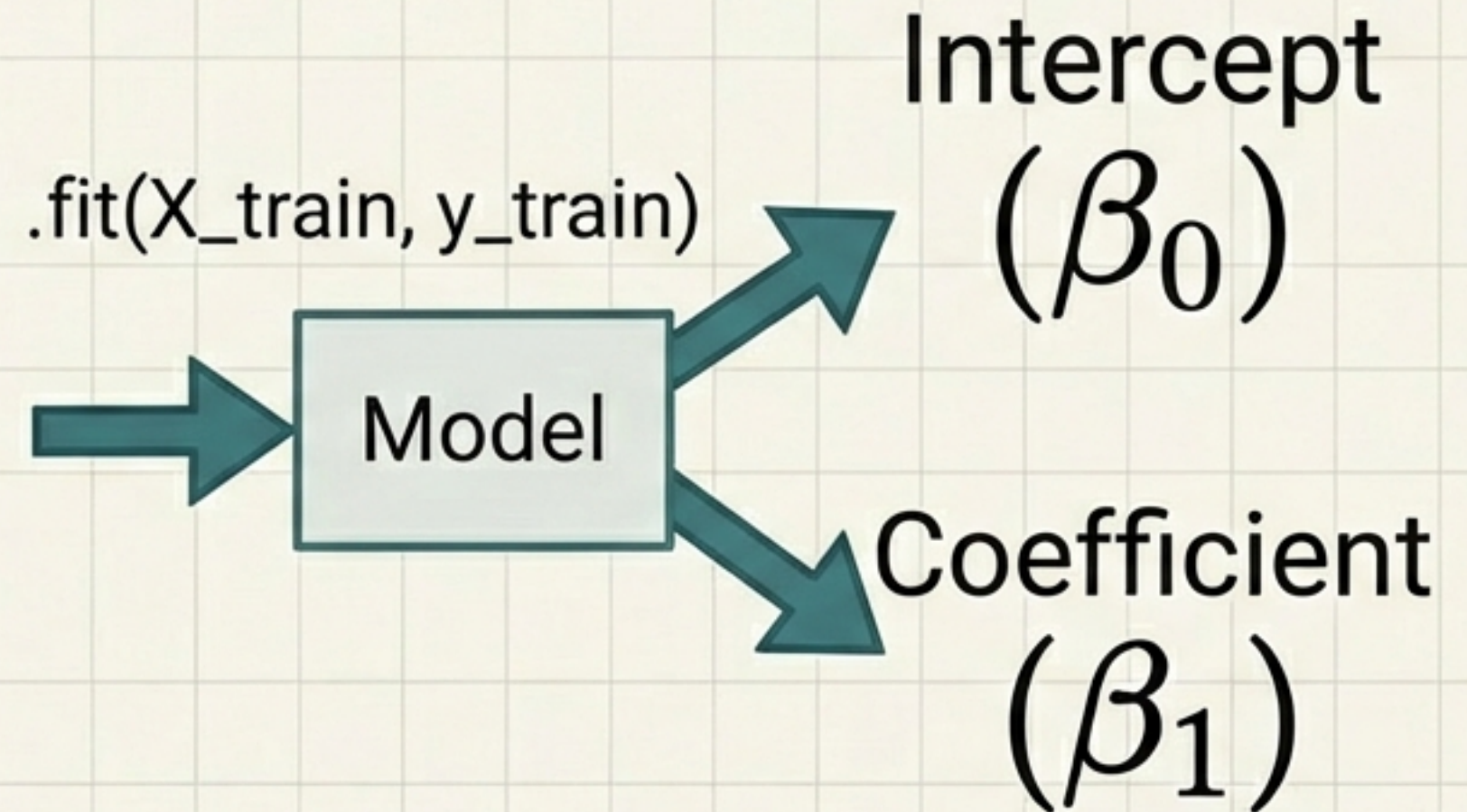
Full Dataset

Study Material
(Train)

Exam
Questions
(Test)

Step 3 & 4: Training the Model

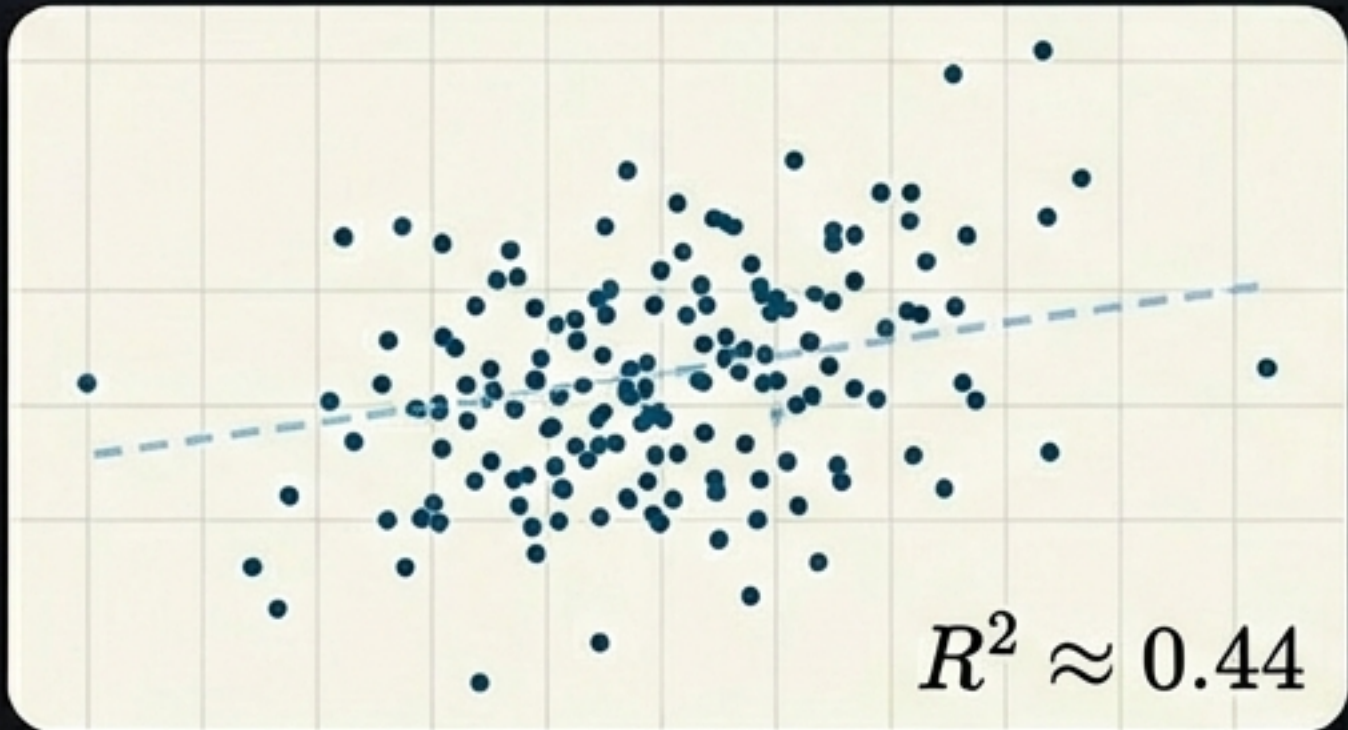
```
model = LinearRegression()  
model.fit(X_train, y_train)  
  
# The machine learns these:  
print(model.intercept_)  
print(model.coef_)
```



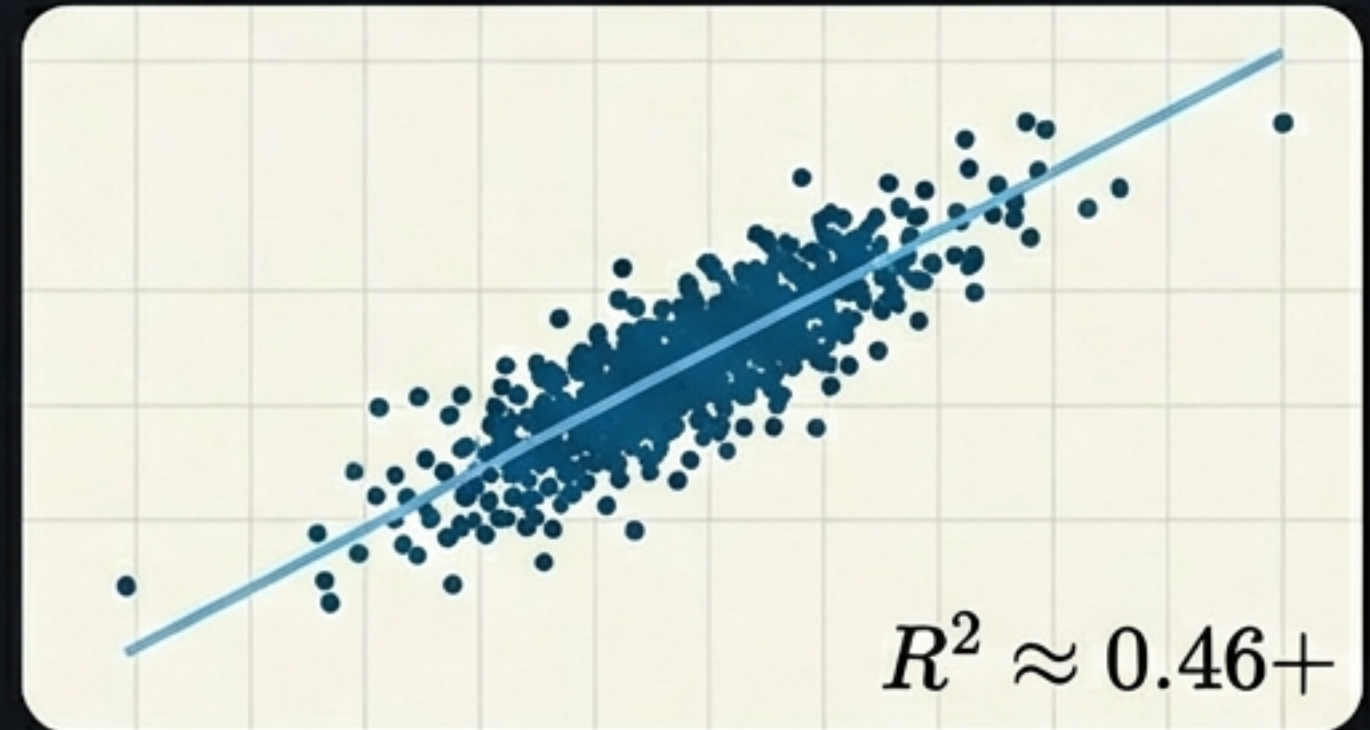
The .fit() command is where the Best Fit Line is drawn.

Optimizing for the Best Fit

Raw Data



Scaled & Cleaned

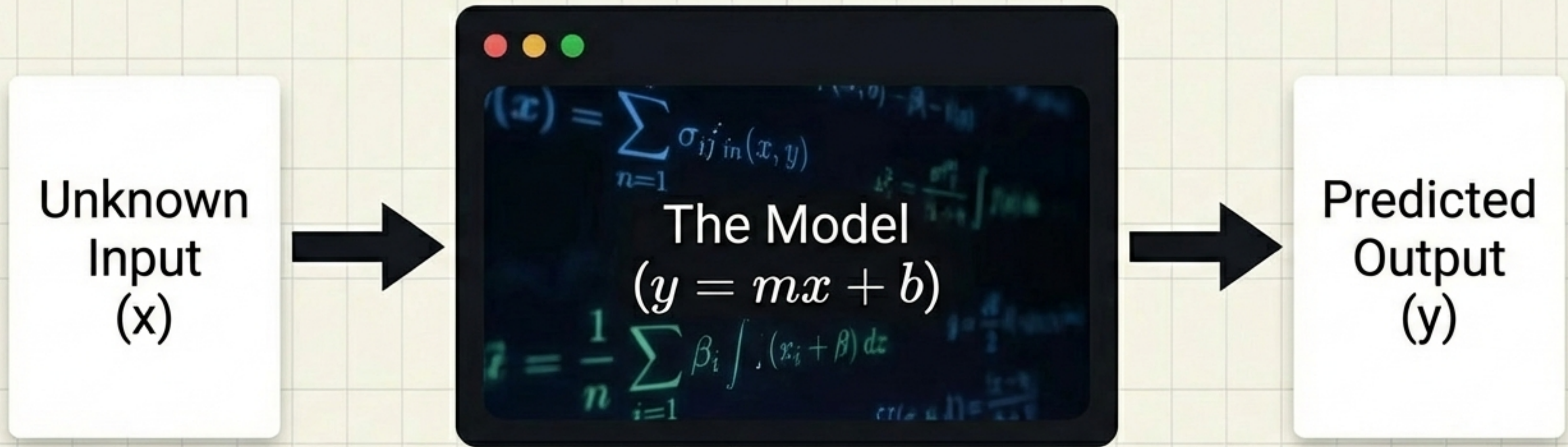


Techniques Used:

1. Standard Scaling (Normalizing Magnitude)
2. Outlier Removal

The model is only as good as the data you feed it.

The Power of Prediction



We have turned historical data into a function that predicts the future.